Speaker 1 - your script

Slides 1-3: introduction, the problem, how the car learns. About 1 min 33 s. Speak slowly, pause at full stops, and look up at the TA between sentences - you don't have to read every word exactly.

Slide 1

[Slide 1 is already on screen. Smile, look at the TA, and start.]

Good afternoon, everyone. We're Bandit LAB. Have you ever stood at a busy crossing, trying to decide when it's safe to go? That's exactly the problem in our paper, “Social Attention for Autonomous Decision-Making in Dense Traffic”, from NeurIPS 2019. It teaches a self-driving car to make that decision.

Slide 2

[Click to slide 2. Point at the picture of the junction.]

Here's the situation. Our car wants to turn left, and there's no traffic light. Cars are coming from every side. If it goes too early, it crashes. If it waits too long, it just sits there forever. We can't write a rule for every possible case, so instead we let the car learn from experience, by trial and error. That's reinforcement learning.

Slide 3

[Click to slide 3. Point at the formula when you say “the formula says”.]

So how does it learn? Every second, the car looks around: where it is, how fast it's going, and the same for up to fourteen cars

nearby. Then it makes a simple choice: slow down, keep its speed, or speed up. If it drives smoothly and quickly, it gets a reward. If it crashes, it loses five points. Over time, the car learns a score, called Q, for each choice. This formula simply says: a good choice now gives a good reward now and leads to a good situation next. Now I'll hand over to my teammate, who'll explain how the car actually learns this score.

[Hand over to Speaker 2.]